

I-BEAMS

STRESS AND DEFLECTION OF BEAMS

No.	Load condition	Bending moment	Reaction force and shear stress	Deflection
1		 $M_x = Px$ $M_{max} = Pa$	 $R_A = P$	 $dc = \frac{Pa^3}{3EI}$ $d_{max} = \frac{Pa^3}{3EI} \left(1 + \frac{3b}{2a}\right)$
2		 $M_{max} = M_x = M_r$	 $R_A = 0$	 $dc = \frac{Ma^2}{2EI}$ $d_{max} = \frac{Ma^2}{2EI} \left(1 + \frac{2b}{a}\right)$
3		 $M_x = \frac{Wx^2}{2a}$ $M_{max} = \frac{Wa}{2}$	 $R_A = W$	 $dc = \frac{wa^3}{8EI}$ $d_{max} = \frac{wa^3}{8EI} \left(1 + \frac{4b}{3a}\right)$
4		 $M_{max} = W\left(a + \frac{b}{2}\right)$	 $R_A = W$	 $d_{max} = \frac{w}{24EI} \times (8a^3 + 18a^2b + 12ab^2 + 3b^3 + 12a^2c + 12abc + 4b^2c)$
5		 $M_x = \frac{Wx^2}{3a^2}$ $M_A = \frac{Wa}{3}$	 $R_A = W$	 $dc = \frac{Wa^3}{15EI}$ $d_{max} = \frac{Wa^3}{15EI} \left(1 + \frac{5b}{4a}\right)$
6		 $M_{max} = W\left(a + \frac{2b}{3}\right)$	 $R_A = W$	 $d_{max} = \frac{W(20a^3 + 50a^2b + 40ab^2 + 11b^3)}{60EI}$
7		 $M_{max} = \frac{Pl}{4}$	 $R_A = R_B = \frac{P}{2}$	 $d_{max} = \frac{Pl^3}{48EI}$
8		 $M_{max} = \frac{Pab}{l}$	 $R_A = Pb/l$ $R_B = Pa/l$	 d_{max} takes place in the range of 0.0774l from the span center. $d_{center} = \frac{Pl^3}{48EI} \left[\frac{3a}{l} - 4\left(\frac{a}{l}\right)^3 \right]$
9		 $M_C = \frac{Pa(b+2c)}{l}$ $M_D = \frac{Pc(b+2a)}{l}$	 $R_A = \frac{P(b+2c)}{l}$ $R_B = \frac{P(b+2a)}{l}$	Deflection of span center is calculated by applying formula given in No.8 to each load.
10		 $M_{max} = \frac{Pl}{3}$	 $R_A = R_B = P$	 $d_{max} = \frac{23Pl^3}{648EI}$

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, in-case of a beam subject to uniformly distributed load, w, W equals $w\ell$, where ℓ is span length.